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Roll No. 32

Batch T12

Experiment No-5

Aim: Experiment 5: To Build the pipeline of jobs using Maven / Gradle / Ant in Jenkins, create a pipeline script to Test and deploy an application over the tomcat server

Programming in Jenkins:

Continuous Integration is a software development practice where members of a team integrate their work frequently, usually each person integrates at least daily leading to multiple integrations per day. Each integration is verified by an automated build (including test) to detect integration errors as quickly as possible.” In simple way, Continuous integration (CI) is the practice of frequently building and testing each change done to your code automatically.

Jenkins is a self-contained, open-source automation server which can be used to automate all sorts of tasks related to building, testing, and delivering or deploying software.

Our first job will execute the shell commands. The freestyle project provides enough options and features to build the complex jobs that you will need in your projects.

Example 1

Example 1.1: Deploying a freestyle app in Jenkins

Creating a job:

Start building your software project



Naming the job and setting it as freestyle:

The screenshot shows a dashboard for a build system. At the top, there's a header with a user profile icon, the word "Dashboard", and the project ID "24_34_31_45_40_42_37_41". Below the header is a sidebar with navigation links: Status, Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area is titled "24_34_31_45_40_42_37_41" and includes a "Permalinks" section with a list of build events: "Last build (#6), 31 min ago", "Last stable build (#6), 31 min ago", "Last successful build (#6), 31 min ago", and "Last completed build (#6), 31 min ago". Below this is a "Builds" section with a search filter and a table of builds. The table has columns for build number, time, and a dropdown menu. The builds listed are #6, #5, #4, #3, #2, and #1, all at 10:37 AM. On the right side of the dashboard, there is a vertical scrollbar and a link to "Add description".

Dashboard 24_34_31_45_40_42_37_41

Status 24_34_31_45_40_42_37_41

Changes

Workspace

Build Now

Configure

Delete Project

Rename

Permalinks

- Last build (#6), 31 min ago
- Last stable build (#6), 31 min ago
- Last successful build (#6), 31 min ago
- Last completed build (#6), 31 min ago

Builds

Filter /

Today

#6	10:41 AM	▼
#5	10:37 AM	▼
#4	10:37 AM	▼
#3	10:37 AM	▼
#2	10:37 AM	▼
#1	10:37 AM	▼

Add description

Selecting build type as “Execute shell”:

Build Steps

The screenshot shows a dropdown menu for "Build Steps". The menu is titled "Add build step ^" and contains a search filter. The list of build steps includes: "Execute Windows batch command", "Execute shell", "Invoke Ant", "Invoke Gradle script", "Invoke top-level Maven targets", "Run with timeout", and "Set build status to 'pending' on GitHub commit". The "Execute shell" option is highlighted.

Add build step ^

Filter

- Execute Windows batch command
- Execute shell
- Invoke Ant
- Invoke Gradle script
- Invoke top-level Maven targets
- Run with timeout
- Set build status to "pending" on GitHub commit

Entering a simple command for the shell execution:

Dashboard24_34_31_45_40_42_37_41Configuration

Configure

General

Source Code Management

Triggers

Environment

Build Steps

Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute Windows batch command

Command

See the list of available environment variables

javac C:\Users\richminds\Desktop\sepm\24.java
java C:\Users\richminds\Desktop\sepm\24.java

Advanced

Add build step

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Add post-build action

Apply

Applying and saving the project configuration:

Save

Apply

✓ Saved

Building the project:

▶ Build Now

Console output (after building):

Dashboard24_34_31_45_40_42_37_41#1Console Output

Status

Changes

Console Output

Edit Build Information

Delete build #1

Timings

Next Build

Console Output

DownloadCopyView as plain text

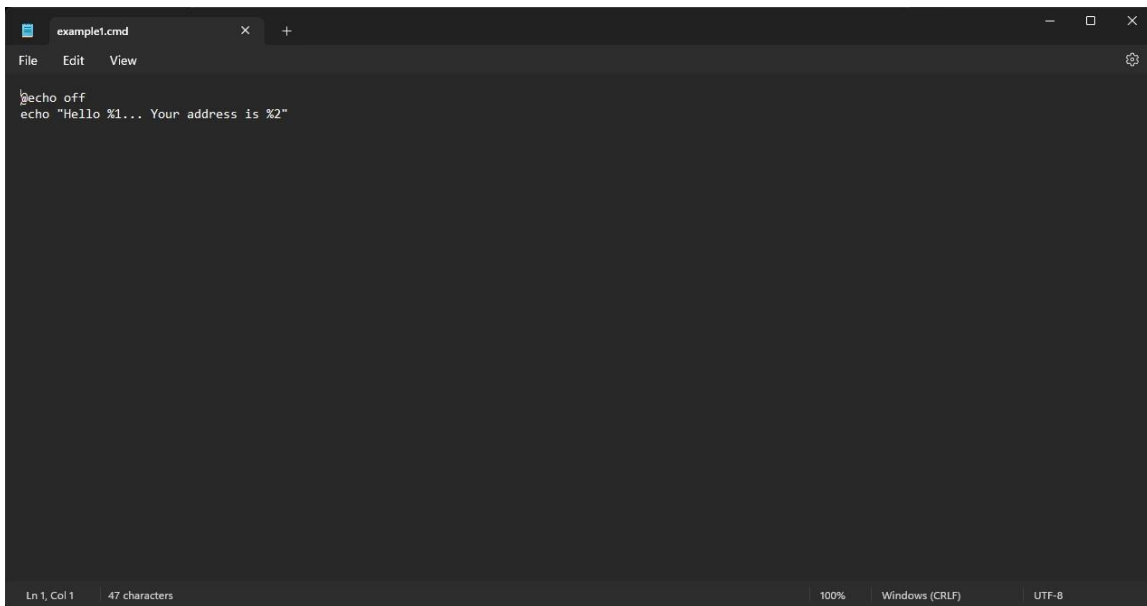
Started by user Aditya Dikonda
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41
[24_34_31_45_40_42_37_41] \$ cmd /c call C:\WINDOWS\TEMP\jenkins3499849351956503998.bat

C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41>javac C:\Users\richminds\Desktop\sepm\24.java

C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41>java C:\Users\richminds\Desktop\sepm\24.java
This is T12

C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41>exit 0
Finished: SUCCESS

REST APIJenkins 2.503

A screenshot of a code editor window titled 'example1.cmd'. The editor has a dark theme and shows the following text:

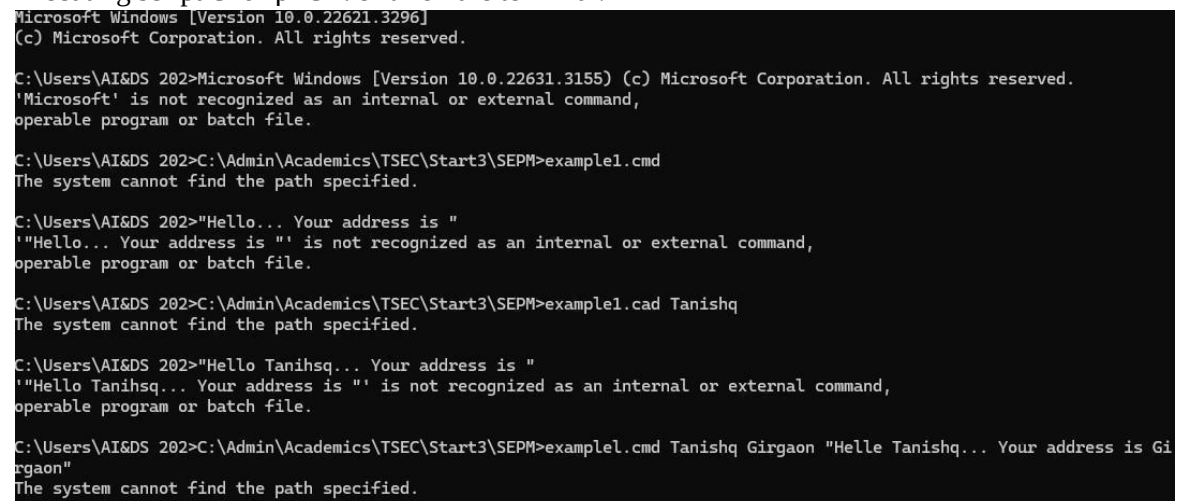
```
echo off
echo "Hello %1... Your address is %2"
```

 The status bar at the bottom indicates 'Ln 1, Col 1', '47 characters', '100%', 'Windows (CRLF)', and 'UTF-8'.

Example 1.2: Taking parameters through files

Contents of script example1.cmd:

Executing script example1.cmd on the terminal:

A screenshot of a Windows terminal window. The output shows several attempts to run the script:

```
Microsoft Windows [Version 10.0.22621.3296]
(c) Microsoft Corporation. All rights reserved.

C:\Users\AI&DS 202>Microsoft Windows [Version 10.0.22631.3155] (c) Microsoft Corporation. All rights reserved.
'Microsoft' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPM>example1.cmd
The system cannot find the path specified.

C:\Users\AI&DS 202>"Hello... Your address is "
'"Hello... Your address is "' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPM>example1.cad Tanishq
The system cannot find the path specified.

C:\Users\AI&DS 202>"Hello Tanishq... Your address is "
'"Hello Tanishq... Your address is "' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\AI&DS 202>C:\Admin\Academics\TSEC\Start3\SEPM>example1.cmd Tanishq Girgaon "Helle Tanishq... Your address is Gi
rgaon"
The system cannot find the path specified.
```

Modifying the Jenkins project to execute the script while supplying required parameters:

A screenshot of the Jenkins 'Build Steps' configuration page. The 'Execute Windows batch command' step is selected. The 'Command' field contains the text:

```
C:\Admin\Academics\TSEC\Start3\SEPM\example1.cmd Siddhant Goregaon
```

 Below the command field is an 'Advanced' dropdown menu. At the bottom of the configuration area is an 'Add build step' button.

Console output after building the modified project:



: Running
a Java program under Jenkins

Creating a simple Java program:

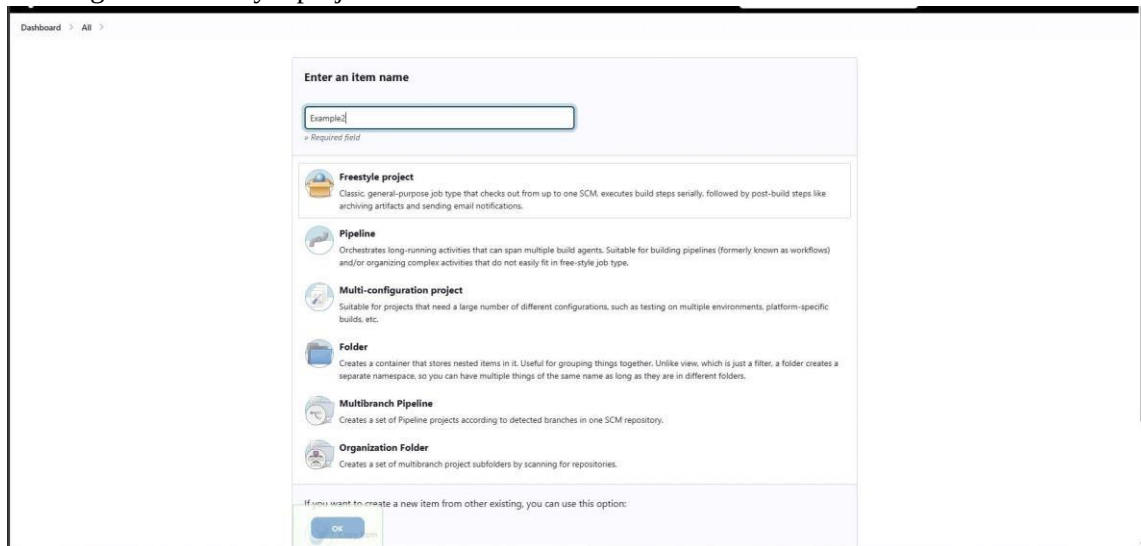
Compiling and running the program on the terminal:

```
C:\Users\richminds\Desktop\sepm>javac 24.java

C:\Users\richminds\Desktop\sepm>java 24.java
This is T12

C:\Users\richminds\Desktop\sepm>
```

Creating a new freestyle project:



Configure new project:

Command

See the list of available environment variables

```
javac C:\Users\richminds\Desktop\sepm\24.java
java C:\Users\richminds\Desktop\sepm\24.java
```

Console output after building:

Console Output

[Download](#)[Copy](#)[View as plain text](#)

```
Started by user Aditya Dikonda
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41
[24_34_31_45_40_42_37_41] $ cmd /c call C:\WINDOWS\TEMP\jenkins3970528995341461278.bat

C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41>javac C:\Users\richminds\Desktop\sepm\24.java

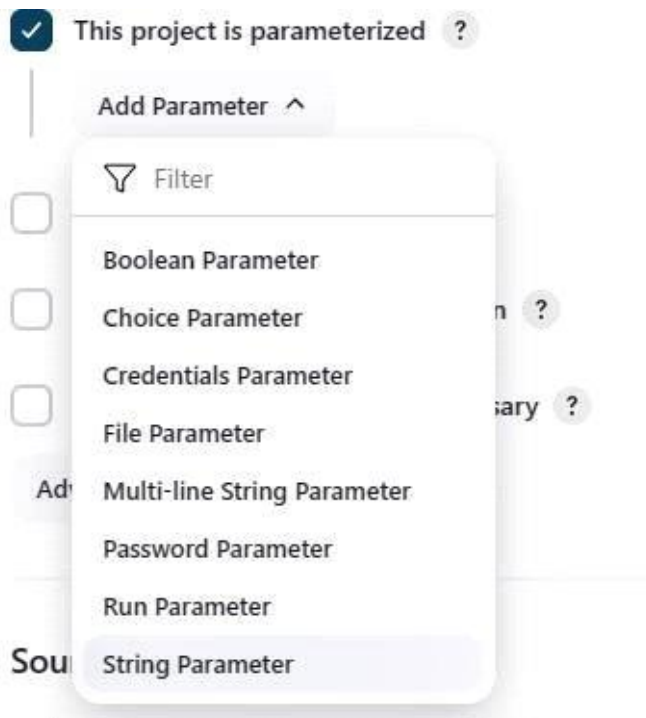
C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41>java C:\Users\richminds\Desktop\sepm\24.java
This is T12

C:\ProgramData\Jenkins\jenkins\workspace\24_34_31_45_40_42_37_41>exit 0
Finished: SUCCESS
```

Example 3

Example 3.1: Parameterise build Creating
a new freestyle project:

Enabling parameterisation and adding a String parameter:



Configuring the string parameter as Fname:

String Parameter ?

×

Name ?

Fname

Default Value ?

Description ?

Plain text [Preview](#)

☐ Trim the string ?

Adding a choice parameter and configuring it as **City** with the following choices:

Choice Parameter

×

Name

City

Choices

Ambernath
Badlapur
Kalyan
Dombivli

! Requires Choices.

Description

Configuring build steps:

Build Steps

Execute Windows batch command ?

×

Command

See [the list of available environment variables](#)

C:\Admin\Academics\TSEC\Start3\SEPM\example3.cmd %Fname% %City%

Advanced ▾

Add build step ▾

Entering parameters for build:

Project Example3

This build requires parameters:

Fname

Siddhant

City

Bandra ▾

▶ Build

Cancel

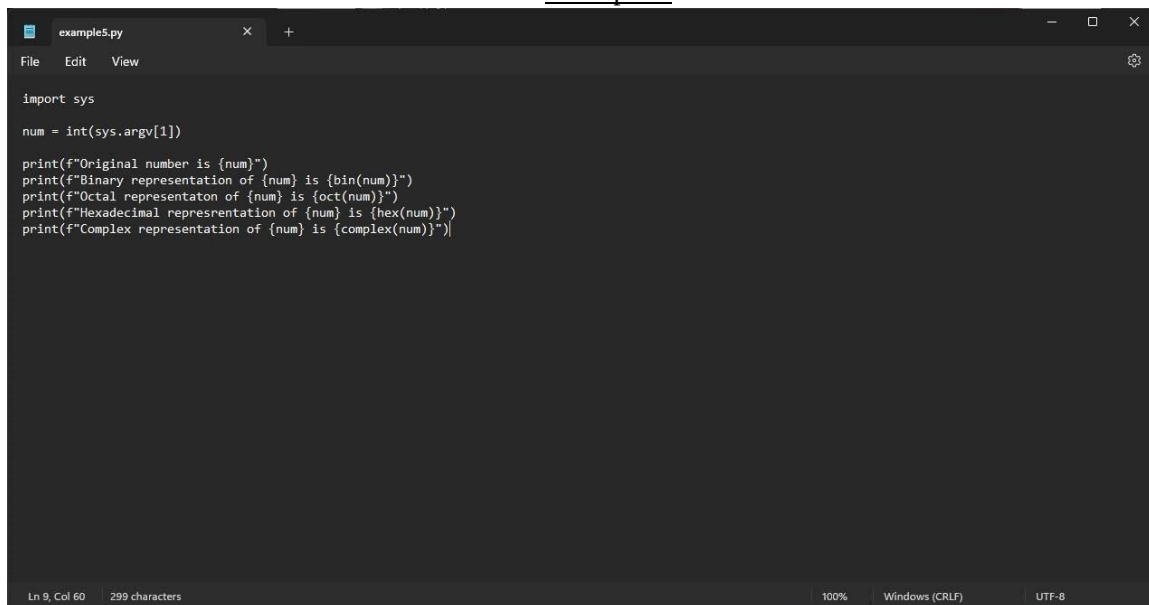
Console output after building:

✓ Console Output

```
Started by user Siddhant Chetlur
Running as SYSTEM
[EnvInject] - Loading node environment variables.
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\Example3
[Example3] $ cmd /c call C:\WINDOWS\TEMP\jenkins14094536165150906151.bat

C:\ProgramData\Jenkins\jenkins\workspace\Example3>C:\Admin\Academics\TSEC\Start3\SEPM\example3.cmd Siddhant Bandra
Hello your name is Siddhant and your city is Bandra
Finished: SUCCESS
```

Example 5



```
example5.py
File Edit View

import sys

num = int(sys.argv[1])

print(f"Original number is {num}")
print(f"Binary representation of {num} is {bin(num)}")
print(f"Octal representation of {num} is {oct(num)}")
print(f"Hexadecimal representation of {num} is {hex(num)}")
print(f"Complex representation of {num} is {complex(num)}")

Ln 9, Col 60 299 characters 100% Windows (CRLF) UTF-8
```

Example 5.1: Running a Python program

Creating a simple Python script:

Running the Python script on the terminal:

```
C:\Users\richminds\Desktop\sepm>python 24.py 10
Number is 10

C:\Users\richminds\Desktop\sepm>_
```

Creating a new freestyle project:

Enter an item name

Example5

» Required field


Freestyle project

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.


Pipeline

Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.


Multi-configuration project

Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.


Folder

Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.


Multibranch Pipeline

Creates a set of Pipeline projects according to detected branches in one SCM repository.


Organization Folder

Creates a set of multibranch project subfolders by scanning for repositories.

If you want to create a new item from other existing, you can use this option:

OK

Parameterising the project with a string parameter as follows:

☒ This project is parameterized ?

String Parameter ?

Name ?

num

Default Value ?

Description ?

Plain text [Preview](#)

☐ Trim the string ?

Add Parameter ▼

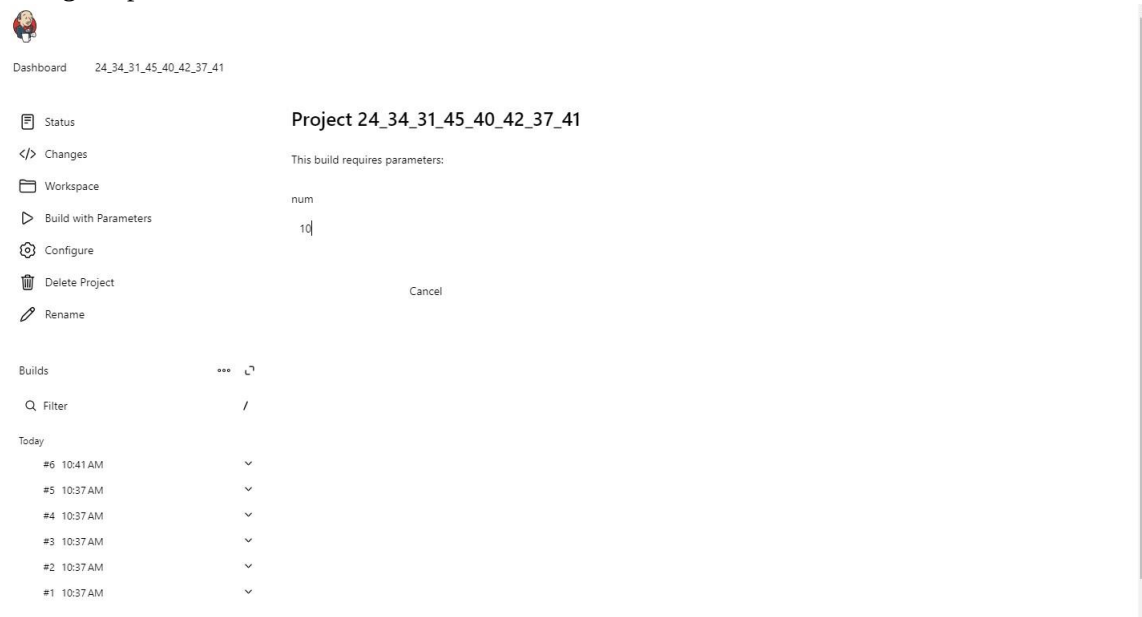
Configuring the build steps:

Command

See the list of available environment variables

```
python C:\Users\richminds\Desktop\sepm\24.py
```

Setting the parameter for the build:



Conclusion: Thus, we have successfully studied Continuous Integration and installed, configured, and understood programming with Jenkins.